

Research Statement

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Methods: DCE, Structural Choice Modeling, LLM-Based Simulation, Behavioral Digital Twins

Research Overview

My research builds computational frameworks that translate behavioral models of patient and clinician decision-making into actionable, human-centered decision-support tools. I bridge behavioral economics, discrete choice methods, and generative AI to create systems that not only predict behavior but can be interacted with, audited, and deployed alongside real users.

The Behavioral Data Toolkit (BDT)

My dissertation developed the **Behavioral Data Toolkit (BDT)**, a three-tier architecture that transforms Discrete Choice Experiment (DCE) data into high-fidelity synthetic agents:

- **Neural-Behavioral Ingestion** — extracts latent preference parameters (risk tolerance, time discounting, trust sensitivity) from multi-modal choice data.
- **Probabilistic Decision Engine** — a theory-agnostic core embedding structural econometric models (Random Utility / Random Regret Minimization) with Bayesian inference, enabling counterfactual simulation.
- **Synthetic Simulation Layer** — deploys thousands of digital agents into virtual policy environments, generating synthetic micro-data for pre-screening interventions before field RCTs.

In a vaccination DCE (N=1,027, Wuhan, China), BDT reduced frontier-deviation MSE by 93.8% compared to raw LLM agents and identified clinically meaningful heterogeneity in delay sensitivity by institutional trust — a pattern conventional DCE analysis alone would not have revealed.

Alignment with Dr. Raj's Group

BDT is currently strongest at *modeling* behavioral decision processes, but the question driving Dr. Raj's group — how to design decision-support tools that patients and clinicians actually trust and use — is exactly where my work needs to go next. I see three concrete connections:

1. **Wearable health data as behavioral input.** BDT's ingestion layer currently handles experimental choice data; extending it to stream from passive sensor data (wearables, EHR logs) would allow preference models to update dynamically — aligning with the group's interest in making continuous health data actionable for chronic condition management.
2. **Red-teaming as validation.** The group's approach of stress-testing LLMs in clinical settings (e.g., red-teaming ChatGPT) provides a natural audit framework for BDT's synthetic agents. Behavioral digital twins must be robust to adversarial inputs before they can be trusted in shared decision-making.
3. **Human-AI interaction design.** BDT currently outputs preference estimates and policy simulations, not interfaces. The group's expertise in designing interaction modalities that support metacognitive engagement — helping users understand when and how to rely on AI recommendations — would transform BDT from a modeling tool into a deployable decision-support system.

Proposed Direction

I propose to spend my postdoc adapting BDT's preference-elicitation framework into an interactive decision-support tool, validated through both behavioral experiments and clinical deployment studies — contributing to the group's mission of making health data and algorithms genuinely actionable for the people who need them.

Selected Publications: Jin, Y. (2026). *Empirical-Frontier Regularization for LLM Synthetic Agents*. SSRN. | Jin, Y. (2026). *Waiting Time as a Behavioral Barrier to Vaccination Uptake*. Job Market Paper.